

지도학습

Learning Strategies: Supervised Learning

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■ 앞으로 2주간...

주차	수업내용	수업방식
1	Lecture Introduction	대면수업
2	Introduction to Pattern Recognition	대면수업
3	Basic of Neural Network → Neural Network의 기본 구조와 동작 방식	대면수업
4	Learning Strategies: Supervised Learning → 지도학습에서의 하이퍼파라미터 튜닝, 규제화, 최적화 방법 학습	대면수업
5	Learning Strategies: Unsupervised Learning → 비지도학습의 동작방식과 학습 방법, 응용 방안 등	대면수업
6	Model Architecture: Convolutional Neural Networks	대면수업
7	Model Architecture: Sequence Models	대면수업
8	Midterm exam	대면수업
9	Visual Anomaly Detection	대면수업
10	Temporal Anomaly Detection	대면수업
11	Predictive Maintenance with Time Series	대면수업
12	Advanced Industrial AI	대면수업
13	Final Project: Proposal	대면수업
14	Final Project: Discussion	대면수업
15	Final Project: Presentation	대면수업

CONTENTS

I

Introduction to Supervised Learning

Supervised Learning Concepts

II

Losses and Logits for Supervised Learning

Type of Classification, Logit and Output Activation, Various Loss Func.

III

Challenging Problems

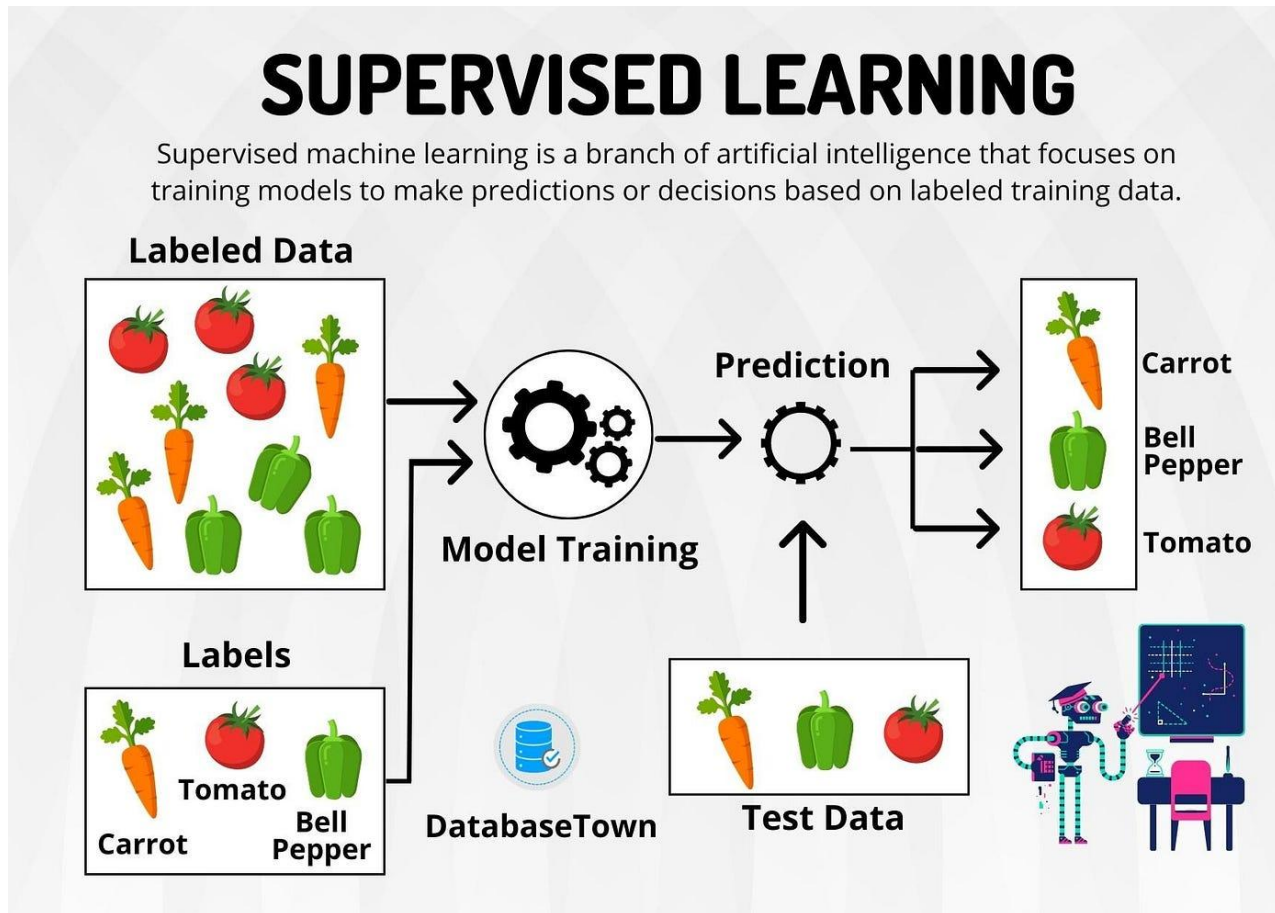
Vanishing Gradient, Overfitting, Imbalanced Data, Data Scarcity

IV

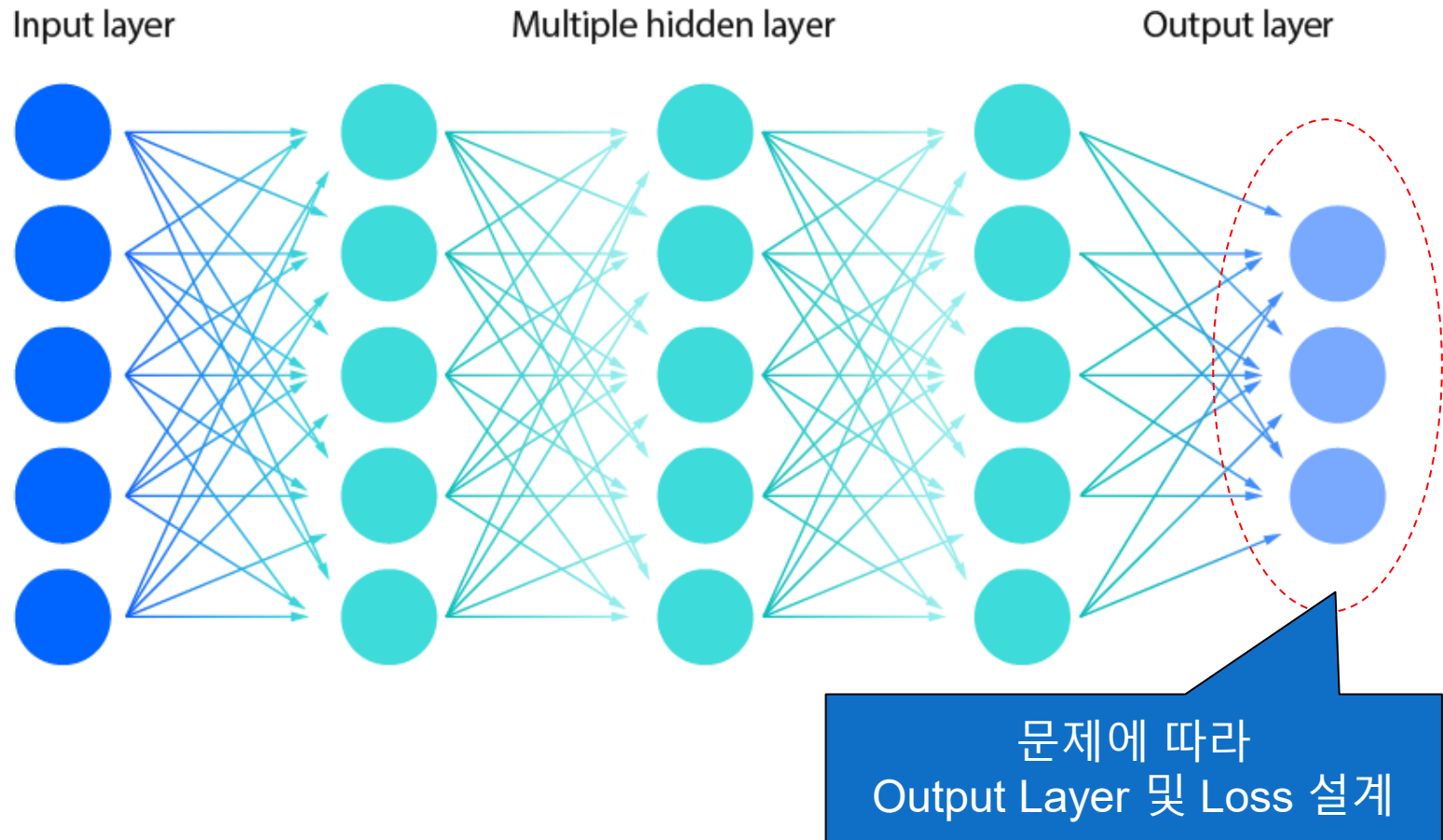
Techniques

Data Augmentation, Normalization, Regularization, Transfer Learning

■ Supervised Learning



■ Supervised Learning

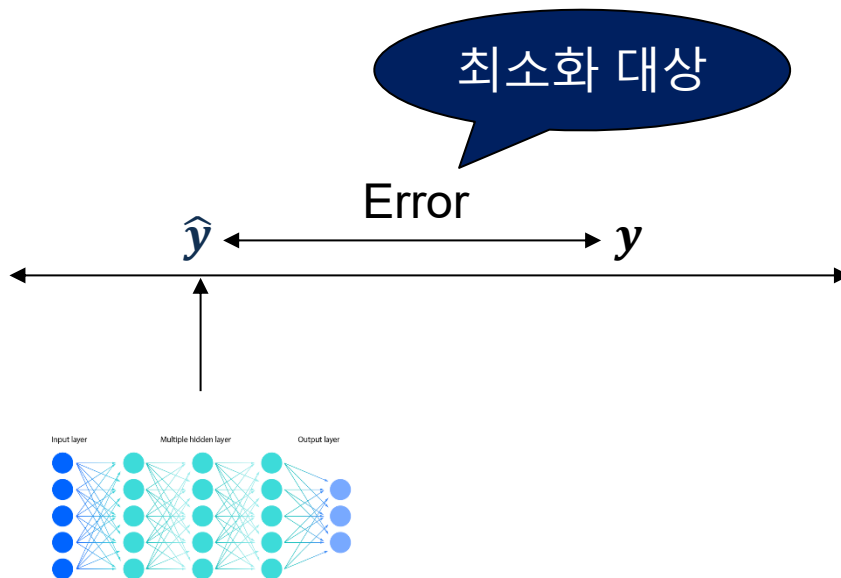


■ Supervised Learning

Input (x)	Output (y)	Type
Home features	Price	Regression
Ad, user Info	Click on ad? (0/1)	Binary classification
Image	Object (1, ..., 1000)	Multi-Class classification
Audio	Text transcript	Seq2seq (classification + alignment)
English	Korean	Seq2seq (classification + alignment)
Image, Radar info	Position of other cars	Multivariate (Multi-output) regression
Movie info.	Genres	Multi-Label classification

■ Loss 란?

- 인공 신경망의 예측 값($\hat{y}$)과 실제 값(y) 간의 차이 (오류)
- 손실 함수 (Cost/Loss Function) 란?
차이(오류)를 정의하는 함수 (ex, Mean Squared Error, Cross Entropy, etc.)



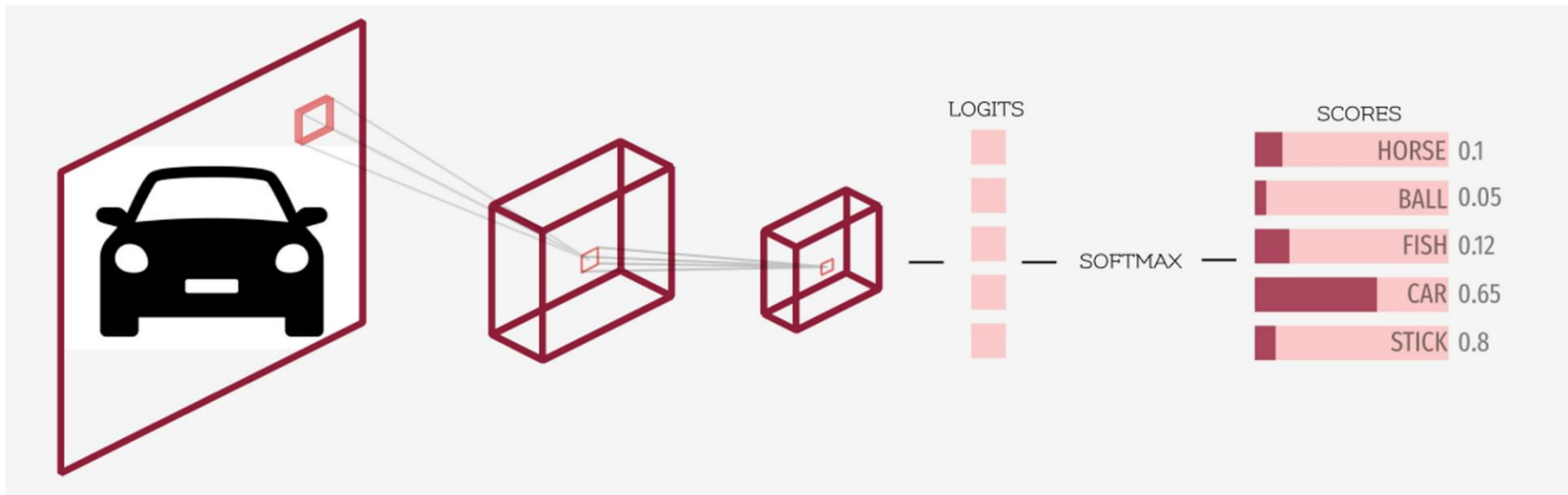
$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

MSE = mean squared error
 n = number of data points
 Y_i = observed values
 $\hat{Y}_i$ = predicted values

$$\text{CE Loss} = - \sum_{i=1}^{\text{output size}} y_i \cdot \log \hat{y}_i$$

■ Logits 이란?

- 신경망의 마지막 선형 계층에서 확률 변환을 적용하기 전의 출력 값
- 즉, 모델이 출력한 가공하지 않은 점수 (raw score)

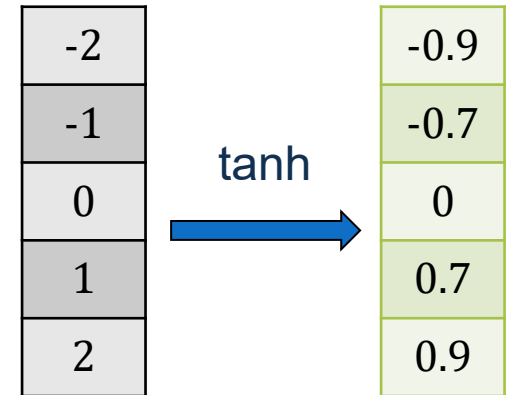
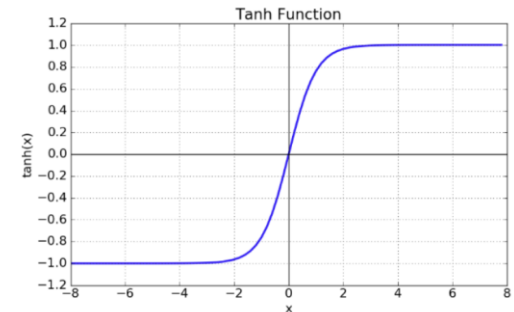
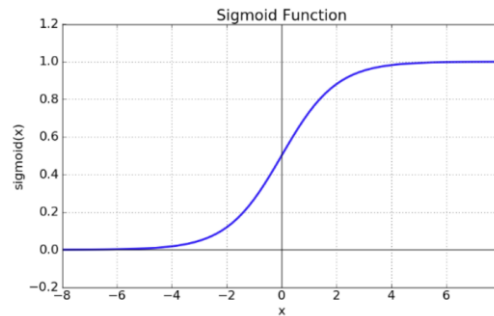
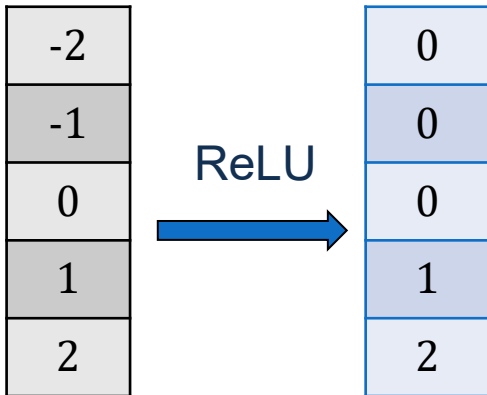
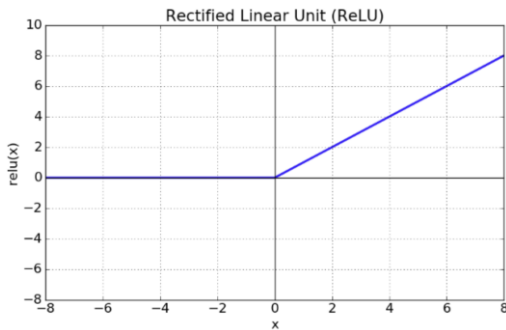


→ 문제에 따라 적합한 출력 활성화함수와 Loss를 설계해야 함!

■ Regression Problem

➤ 출력 활성화 함수: 일반적으로 없음 (identity function)

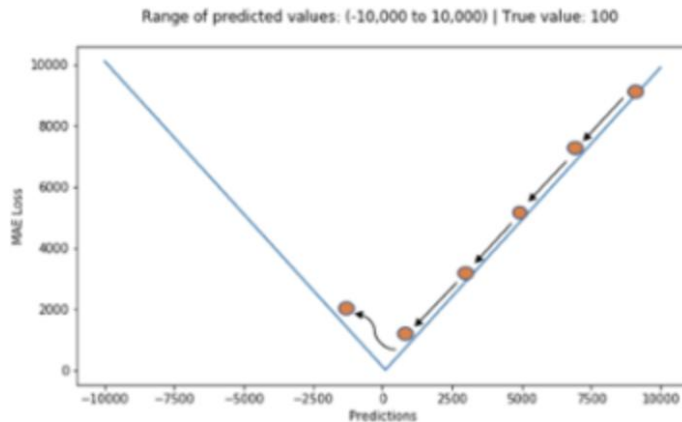
→ 출력 값의 범위를 지정해야 하는 경우 ReLU나 sigmoid, tanh 등 사용 가능



■ Regression Problem

- Loss Function: MAE (Mean Absolute Error, L1 Loss)

$$\frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$



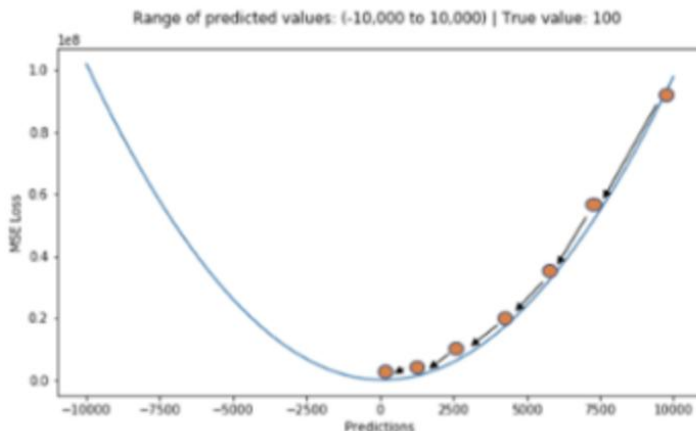
- 이상치에 강건함
→ 일부 노이즈가 있더라도
안정적 학습 가능
- 작은 오차에는 민감하지
않으므로 정밀 예측에서 한계
- 특히, 모든 구간에서 gradient 가
동일하므로, 학습 수렴이 어려움

■ Regression Problem

- Loss Function: MSE (Mean Squared Error, L2 Loss)

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

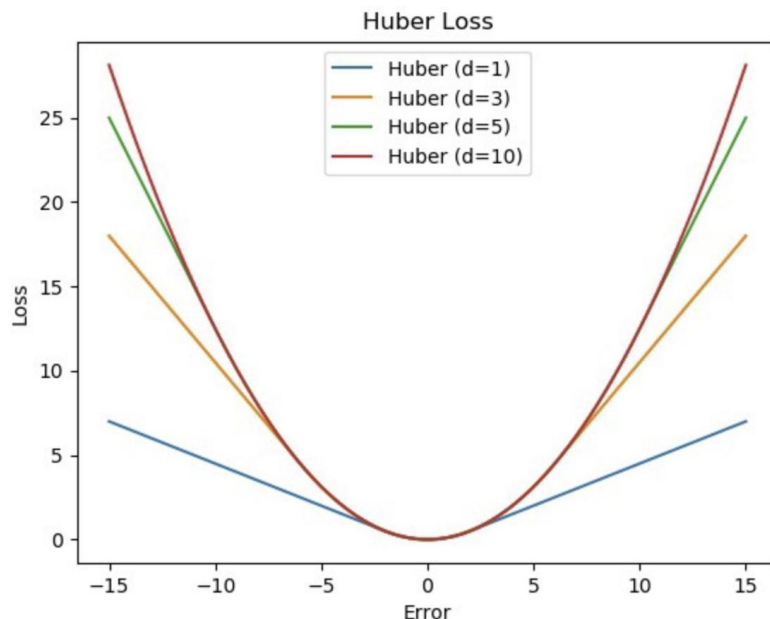
- 작은 오차에 민감함
→ 제곱 항으로 인해 작은 오차가 학습에 영향을 미침 (유용함)
- 이상치에 취약함
→ 제곱 항으로 인해 이상치가 극대화되어 전체 손실을 크게 만들 수 있음
- 함수가 convex하므로, gradient가 달라지기 때문에 수렴에 용이



■ Regression Problem

- Huber Loss (MAE와 MSE의 절충안)

$$L_{\delta}(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{if } |y - \hat{y}| \leq \delta \\ \delta \left(|y - \hat{y}| - \frac{1}{2}\delta \right) & \text{if } |y - \hat{y}| > \delta \end{cases}$$



- MAE의 장점과 MSE의 장점 결합
- 계산 비용 높음 (δ)
- 하이퍼파라미터 고려 (δ)

■ Regression Problem

➤ 실습

[1.nn_regression_loss_comparison.ipynb](#)

■ Classification Problem (Binary Classification)

- 출력 차원 : 1개 class
- 출력 활성화 함수 : 일반적으로 sigmoid
- 손실 함수 : Binary Cross-Entropy (BCE), Log loss라고도 함

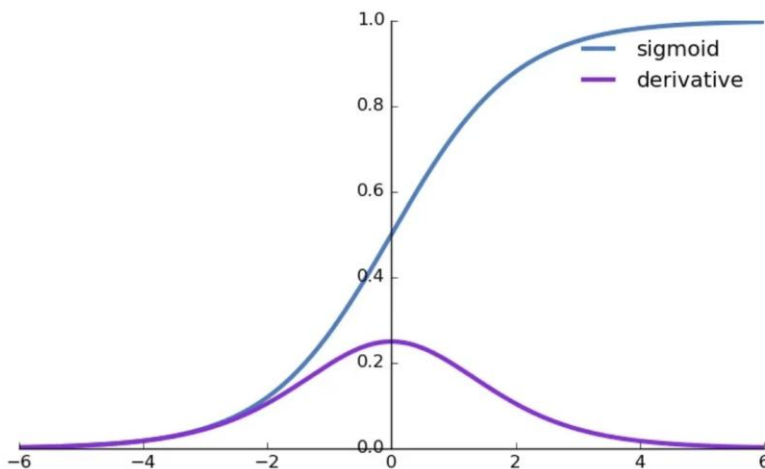
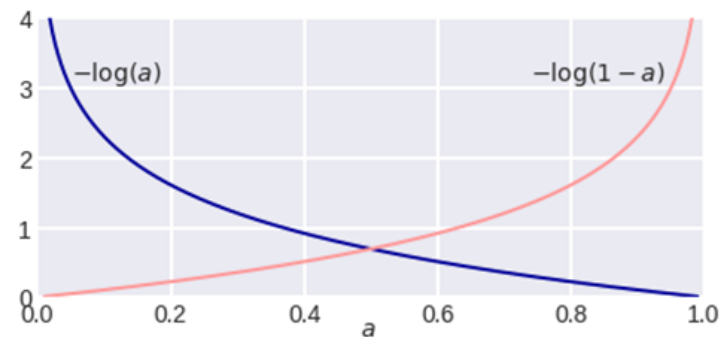


Figure 15. Sigmoid Function Graph Visualization

$$L_{BCE} = y \log \hat{y} + (1 - y) \log(1 - \hat{y})$$

$$-\begin{cases} \log a_i, & y_i = 1, \\ \log(1 - a_i), & y_i = 0. \end{cases}$$

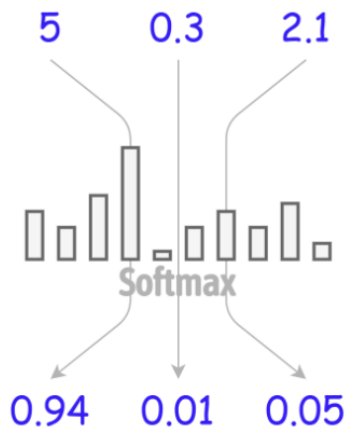


■ Classification Problem (Multi-Class Classification)

- 출력 차원 : K개 class
- 출력 활성화 함수 : 일반적으로 softmax
- 손실 함수 : Categorical Cross Entropy (CE)

$$\text{softmax}(y_i) = \frac{e^{y_i}}{\sum_{i=0}^n e^{y_i}}$$

$$L_{CE} = - \sum_{k=1}^K y_k \log(\hat{y}_k)$$



- CE는 BCE의 일반화된 버전
- $K=2 \rightarrow K>2$

■ Classification Problem (Multi-Label Classification)

- 출력 차원 : K개 class
- 출력 활성화 함수 : 일반적으로 sigmoid
- 손실 함수 : Binary Cross Entropy (BCE)

$$L_{BCE} = -\frac{1}{K} \sum_{k=1}^K (y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k))$$

- 각 Class 별로 구한 BCE를 Class에 대해 평균

■ Classification Problem

➤ 실습

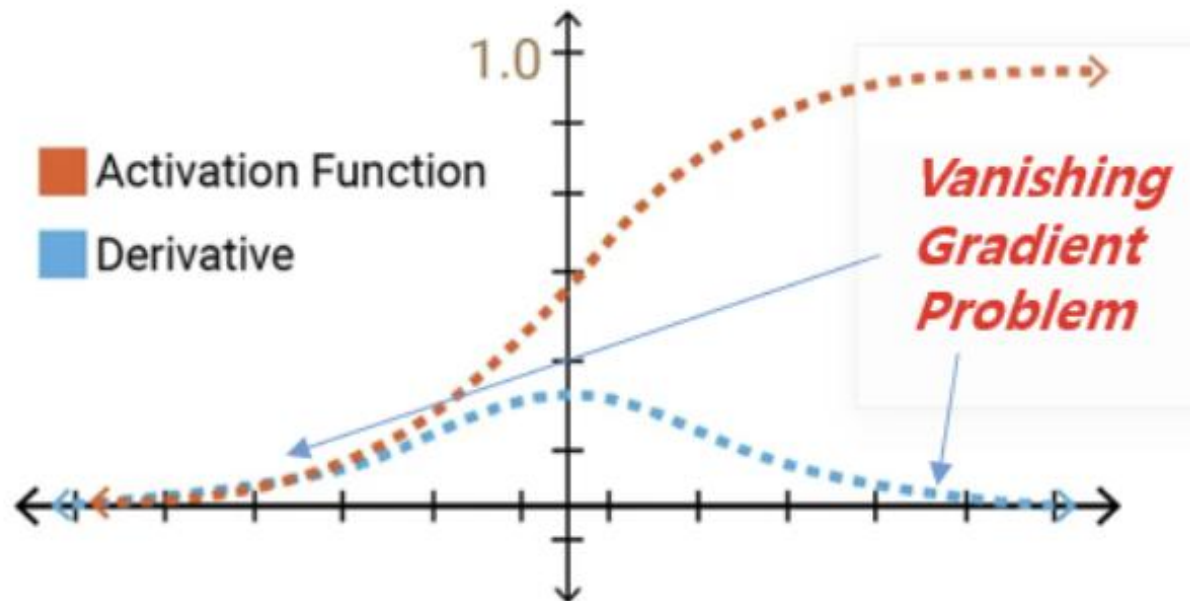
[2.nn_classification_examples.ipynb](#)

[2-1.nn_multiclass_classification.ipynb](#)

■ Challenging Problems : Vanishing Gradient

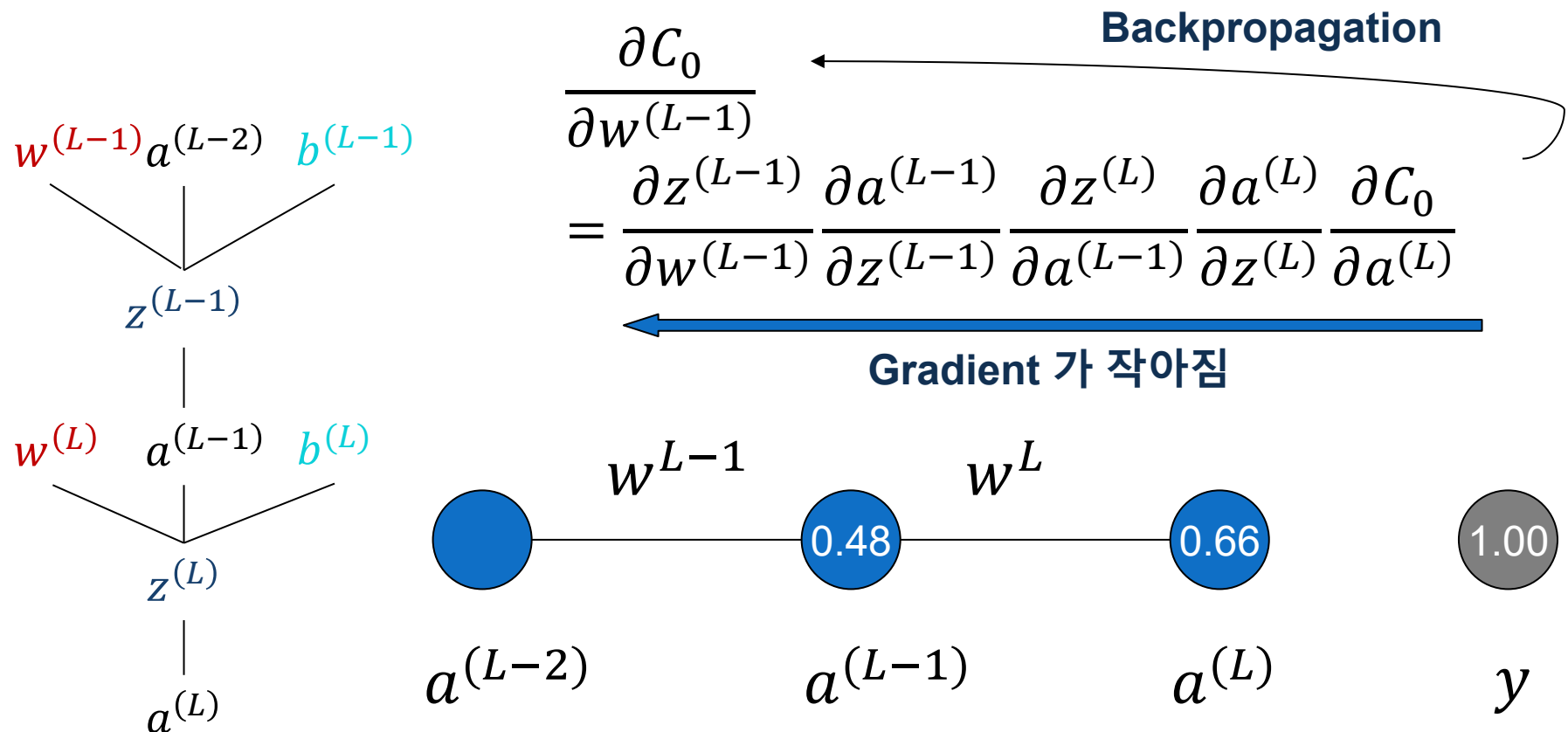
➤ 기울기 소실 (Vanishing Gradient) 문제

Sigmoid Activation Function



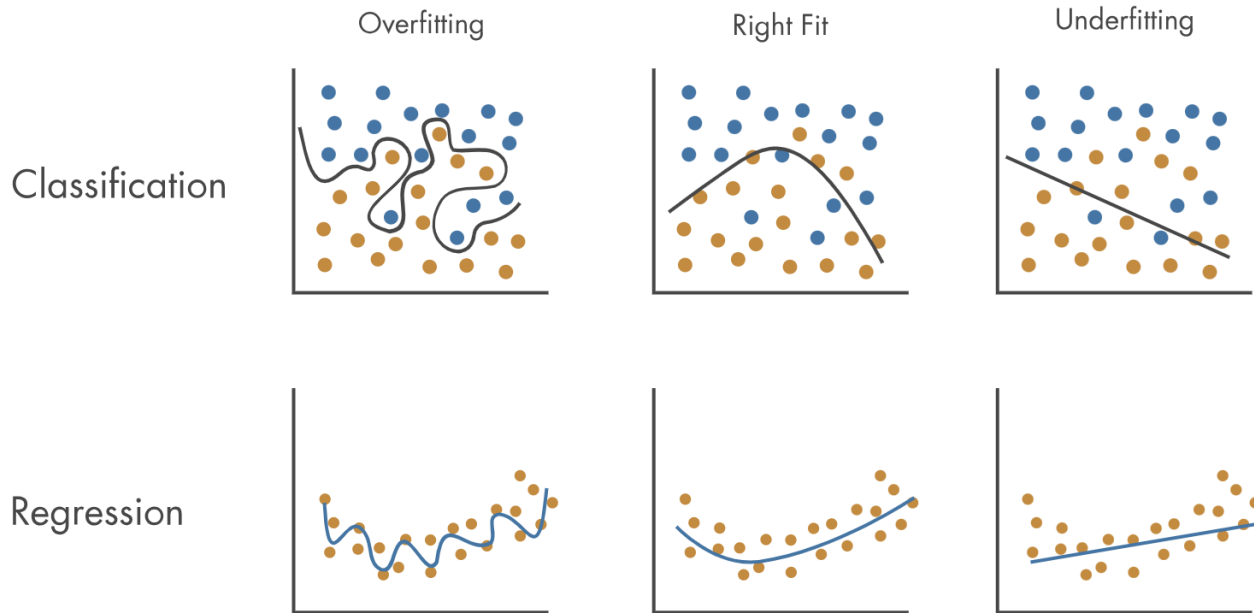
■ Remind & Vanishing Gradient Problem

- 오류 역전파 (Backpropagation) Simple Example
- Chain Rule



■ Challenging Problems : Overfitting

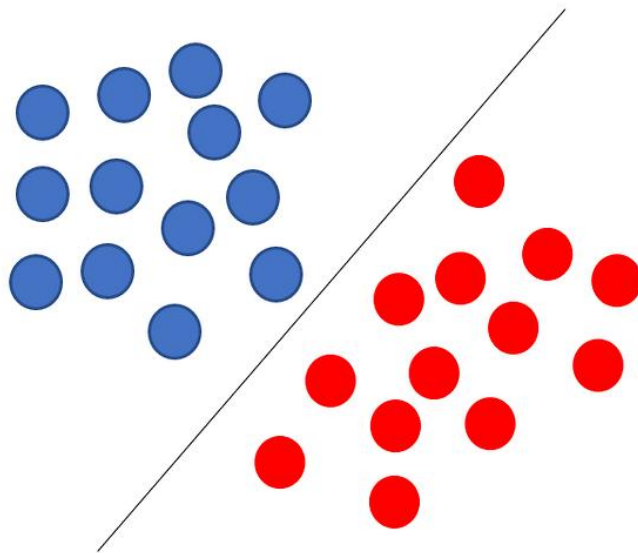
- 딥러닝에서 학습이란 $y = f(x)$ 를 만족하는 $f()$ 를 찾는 것
- 이때, 학습된 모델 $f()$ 가 새로운 데이터(학습데이터x)로 일반화되지 않는 경우 모델이 과적합되었다고 함



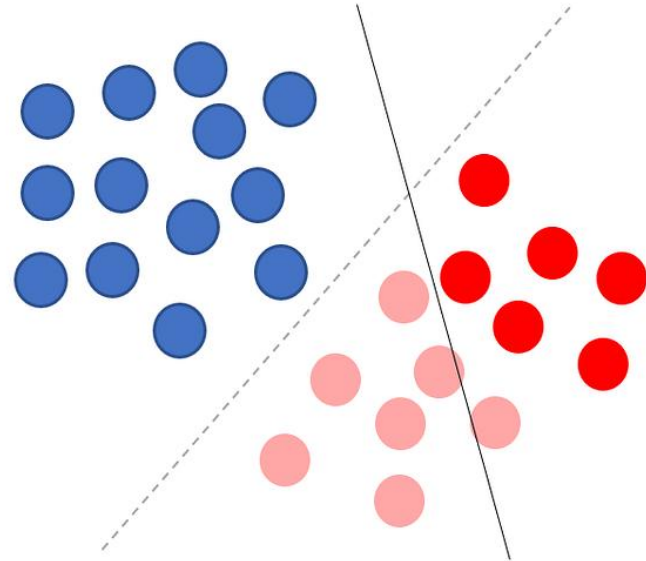
- 과적합 문제가 발생하는 이유1: 학습 데이터의 다양성 부족
- 과적합 문제가 발생하는 이유2: 모델의 복잡도 (파라미터의 양)

■ Challenging Problems : Class Imbalanced

➤ 클래스 불균형 (Class Imbalanced) 문제



Classifier with balanced class



Classifier with imbalanced class

■ Challenging Problems : Data Scarcity

➤ 데이터 부족 (Data Scarcity) 문제

항공/우주



제조



사이버보안



재해/재난



- 데이터가 매우 적음 (1~10개 샘플)
- 데이터가 존재하지 않음 (0개...)
- 또는 정상 데이터만 존재함 (normal : n, abnormal : 0)

■ Challenging Problem

➤ 실습

3.vanishing_gradient_demo.ipynb

4-1.overfitting_dataset_size_demo.ipynb

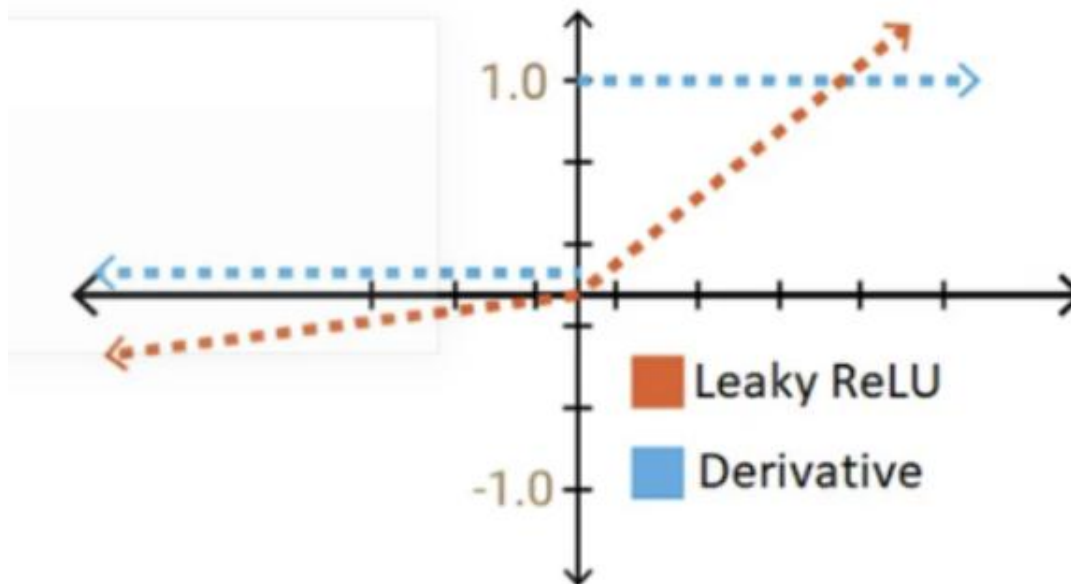
4-2.overfitting_demo.ipynb

5.class_imbalanced_demo.ipynb

■ Challenging Problems : Vanishing Gradient

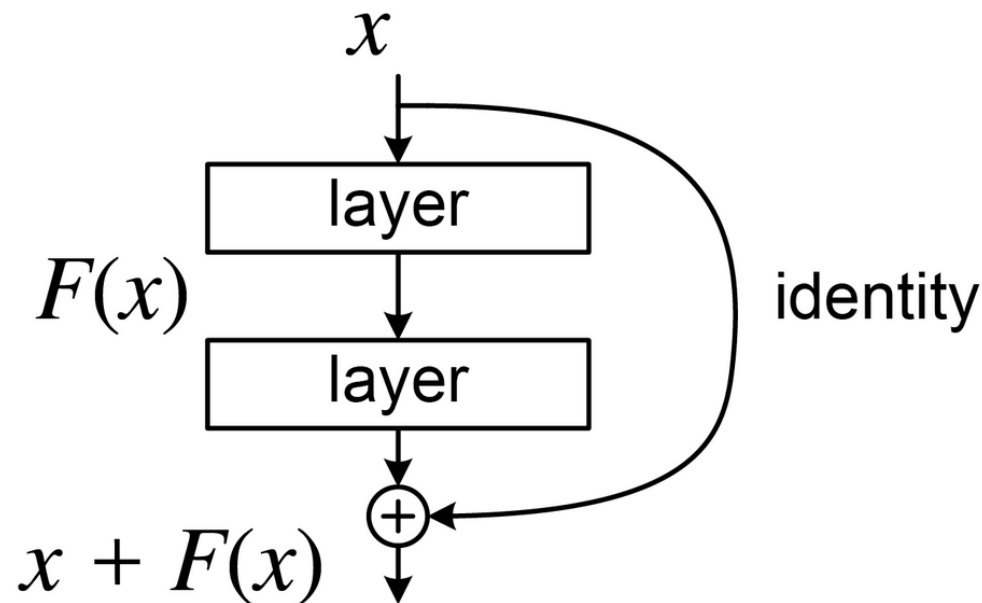
- 기울기 소실 (Vanishing Gradient) 문제
- Solutions : LeRU, Residual

Leaky ReLU Activation Function



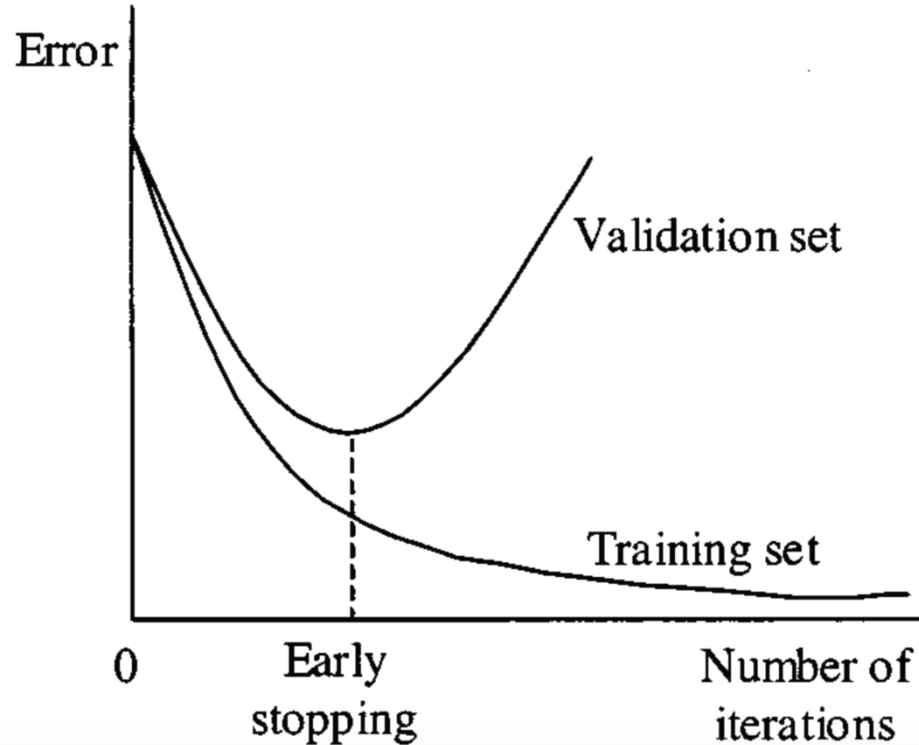
■ Challenging Problems : Vanishing Gradient

- 기울기 소실 (Vanishing Gradient) 문제
- Solutions : LeRU, Residual



■ Challenging Problems : Overfitting

- 과적합 (Overfitting) 문제
- Solutions : Early Stopping, Regularization (L1, L2), Dropout



■ Challenging Problems : Overfitting

- 과적합 (Overfitting) 문제
- Solutions : Early Stopping, Regularization (L1, L2), Dropout

L1 Regularization

$$J_{L1}(W) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^d |w_j|$$

$$\frac{\partial J_{L1}}{\partial w_j} = \frac{1}{N} \sum_{i=1}^N -2(y_i - \hat{y}_i)x_{ij} + \lambda \cdot \text{sign}(w_j)$$

$$w \leftarrow w - \eta \frac{\partial L}{\partial w}$$

■ Challenging Problems : Overfitting

- 과적합 (Overfitting) 문제
- Solutions : Early Stopping, Regularization (L1, L2), Dropout

L2 Regularization

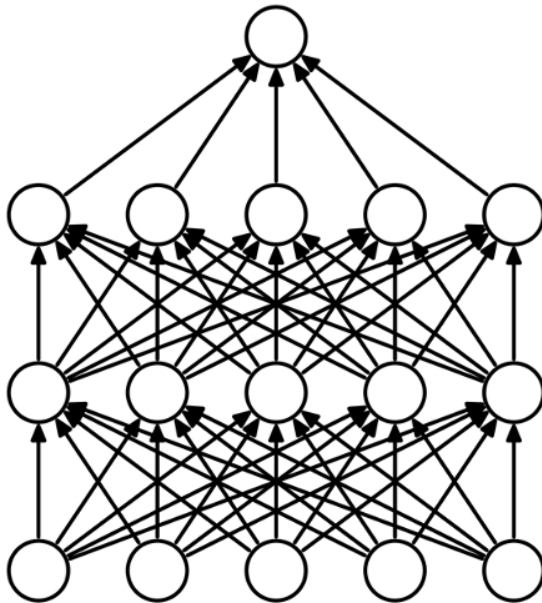
$$J_{L2}(W) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 + \frac{\lambda}{2} \sum_{j=1}^d w_j^2$$

$$\frac{\partial J_{L2}}{\partial w_j} = \frac{\partial}{\partial w_j} \left[\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \right] + \frac{\partial}{\partial w_j} \left[\frac{\lambda}{2} \sum_{j=1}^d w_j^2 \right] = \frac{1}{N} \sum_{i=1}^N -2(y_i - \hat{y}_i)x_{ij} + \lambda w_j$$

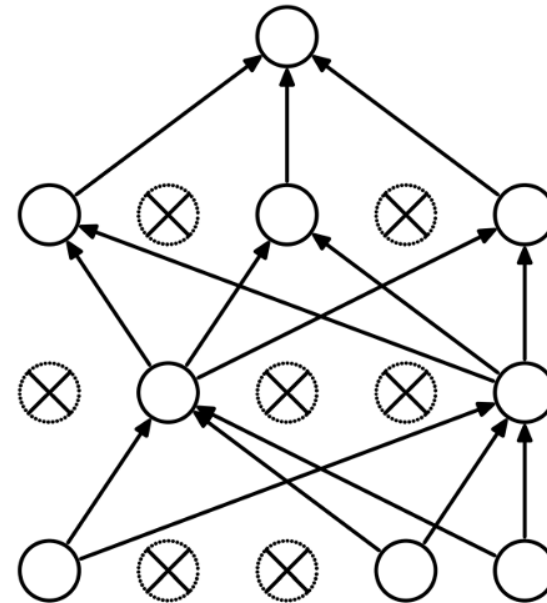
$$w \leftarrow w - \eta \frac{\partial L}{\partial w}$$

■ Challenging Problems : Overfitting

- 과적합 (Overfitting) 문제
- Solutions : Early Stopping, Regularization (L1, L2), Dropout



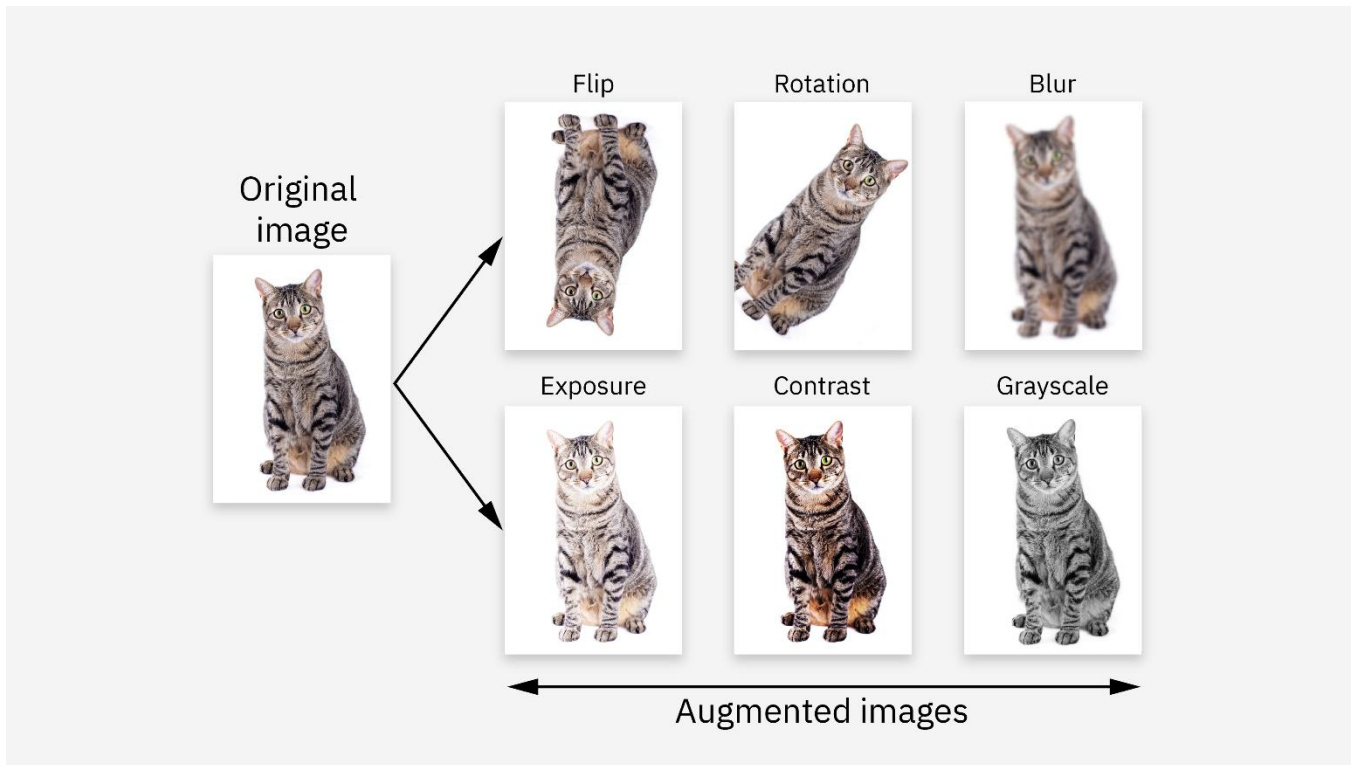
(a) Standard Neural Net



(b) After applying dropout.

■ Challenging Problems : Class Imbalanced

- 클래스 불균형 (Class Imbalanced) 문제
- Solutions : Data Augmentation, Class Weighting)



■ Challenging Problems : Class Imbalanced

- 클래스 불균형 (Class Imbalanced) 문제
- Solutions : Data Augmentation, Class Weighting)

- 소수 클래스에 더 큰 **weight**를 부여, 다수 클래스에는 작은 weight 부여
- 손실 함수 계산 시 각 샘플에 weight 곱함:

$$L = \frac{1}{N} \sum_{i=1}^N w_{y_i} \cdot \ell(\hat{y}_i, y_i)$$

- w_{y_i} : 샘플 i 가 속한 클래스 y_i 의 weight
- ℓ : 일반적인 loss (예: Cross-Entropy)

■ Challenging Problems : Data Scarcity

➤ 데이터 부족 (Data Scarcity) 문제에 대한 솔루션

Data Expansion

- Data Augmentation
- Synthetic Data Generation
- Active Learning

Knowledge Transfer

- Transfer Learning
- Domain Adaptation

Weakly Supervision

- Semi-supervised learning
- Self-supervised Learning
- Unsupervised Learning

Meta Learning

- Few-Shot / One-Shot Learning
- Meta Learning
- Metric Learning

■ Techniques

➤ 실습

[6.residual_vs_plain_demo.ipynb](#)

[7.l1_l2_regularization_demo.ipynb](#)

[8.dropout_demo.ipynb](#)

[9.data_aug_class_weight_demo.ipynb](#)

■ 차후 수업 진행 안내

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감사합니다

Q&A

